Recording L01-R01
Gaussian conditioning by block elimination

A normalized density proof of the Gaussian conditioning formulas

Probabilistic Machine Learning 2027-1

2026-10-17

From a joint Gaussian to a conditional density

From the lecture L01 live lecture: conditioning mean and covariance; independent residual

$$m(x_2) = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

To prove here

$$p(x_1, x_2) = \mathcal{N}(x|\mu, \Sigma)$$

$$p(x_1|x_2) = \mathcal{N}(x_1|m(x_2), S)$$

Assume $\Sigma = \Sigma^T \succ 0$ is symmetric positive definite: $v^T\Sigma v > 0$ for $v \neq 0$. Here $x_i, \mu_i \in \mathbb{R}^{d_i}$ and $d = d_1 + d_2$.

The joint density has a quadratic and a normalizing factor

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$p(x_1, x_2) = \frac{\exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right]}{(2\pi)^{\frac{d}{2}}\sqrt{\det \Sigma}}$$

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$$p(x_1, x_2) = \frac{\exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right]}{(2\pi)^{\frac{d}{2}}\sqrt{\det \Sigma}}$$

$$Q = (x-\mu)^T \Sigma^{-1}(x-\mu), \quad C = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \Sigma}}$$

$$p(x_1, x_2) = C \exp\left(-\frac{Q}{2}\right)$$

> Q controls the exponential; C is the factor that makes the density integrate to one. A conditional density requires both parts, with their correct constants.

The observed covariance block is invertible

Premise Orientation

$$\Sigma = \Sigma^T \succ 0$$

$$z \in \mathbb{R}^{d_2}, \quad z \neq 0, \quad \begin{pmatrix} 0 \\ z \end{pmatrix} \neq 0$$

$$z^T \Sigma_{22} z = \begin{pmatrix} 0 \\ z \end{pmatrix}^T \Sigma \begin{pmatrix} 0 \\ z \end{pmatrix} > 0$$

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$$\Sigma_{22} = \Sigma_{22}^T \succ 0 \quad \Rightarrow \quad \Sigma_{22}^{-1} \text{ exists}$$

Positive definiteness of the joint covariance also holds on the observed coordinate subspace.

Choose the row subtraction to cancel the cross block

First block row minus B times the second

$$T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}, \quad T\Sigma = \begin{pmatrix} \Sigma_{11} - B\Sigma_{21} & \Sigma_{12} - B\Sigma_{22} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

A block diagonal covariance has a block diagonal inverse. In those transformed coordinates, the quadratic separates into two block contributions.

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Choose B to make the upper-right block zero

$$\Sigma_{12} - B\Sigma_{22} = 0$$

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

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The matching column subtraction leaves a block diagonal matrix

First block column minus the second times B^T

$$T\Sigma T^T = \begin{pmatrix} \Sigma_{11} - B\Sigma_{21} & 0 \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

$$\Sigma_{22} B^T = \Sigma_{21}$$

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$$\Sigma_{22} B^T = \Sigma_{21}$$

$$T\Sigma T^T = \begin{pmatrix} \Sigma_{11} - B\Sigma_{21} & 0 \\ \underbrace{\Sigma_{21} - \Sigma_{22} B^T}_{=0} & \Sigma_{22} \end{pmatrix} = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

The remaining first block is the Schur complement of Σ_{22} in Σ, the same residual covariance obtained in the live lecture.

The remaining covariance block is positive definite

Elimination Previous slide

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

$$a \in \mathbb{R}^{d_1}, \quad a \neq 0, \quad T^T \begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ -B^T a \end{pmatrix} \neq 0$$

$$a^T S a = \begin{pmatrix} a \\ 0 \end{pmatrix}^T T\Sigma T^T \begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} a \\ -B^T a \end{pmatrix}^T \Sigma \begin{pmatrix} a \\ -B^T a \end{pmatrix} > 0$$

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$$S = S^T \succ 0 \quad \Rightarrow \quad S^{-1} \text{ exists}, \quad \det S > 0$$

> The residual covariance is a valid nondegenerate Gaussian covariance. Both covariance blocks now have inverses and positive determinants.

The same subtraction produces the centered residual

$$u = x_2 - \mu_2, \quad r = x_1 - \mu_1 - Bu = x_1 - m(x_2)$$

$$T(x - \mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

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$$T(x-\mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

$$T^{-1} = \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}, \quad T^{-1} \begin{pmatrix} r \\ u \end{pmatrix} = \begin{pmatrix} r + Bu \\ u \end{pmatrix} = x - \mu$$

$$TT^{-1} = \begin{pmatrix} I & B - B \\ 0 & I \end{pmatrix} = I = T^{-1}T$$

> u is fixed when conditioning; r is the remaining centered variation around the observation-dependent mean.

The eliminated covariance gives the inverse in the density

Covariance identity Slide 6; T is invertible and both diagonal blocks are positive definite

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

Invert both sides; reverse the order of factors

$$(T^T)^{-1}\Sigma^{-1}T^{-1} = \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix}$$

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Covariance identity Slide 6; T is invertible and both diagonal blocks are positive definite

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

Invert both sides; reverse the order of factors

$$(T^T)^{-1}\Sigma^{-1}T^{-1} = \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix}$$

Multiply on the left by T^T and on the right by T

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} T$$

The recovered inverse covariance is the matrix in Q. Its factor order lets the subtraction act directly on the centered data vector.

Use the eliminated inverse in the quadratic

Inputs Joint density and previous slide

$$Q = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} T$$

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$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} T$$

Substitute the inverse; $(Tv)^T = v^T T^T$

$$Q = [T(x-\mu)]^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} [T(x-\mu)]$$

> The outer factors T act on the centered vector; the middle matrix is block diagonal.

Block diagonal multiplication separates the quadratic

Inputs Previous slide and residual coordinates

$$Q = [T(x-\mu)]^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} [T(x-\mu)]$$

$$T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}, \quad r = x_1 - m(x_2), \quad u = x_2 - \mu_2$$

Block diagonal multiplication separates the quadratic

Inputs Previous slide and residual coordinates

$$Q = [T(x-\mu)]^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} [T(x-\mu)]$$

$$T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}, \quad r = x_1 - m(x_2), \quad u = x_2 - \mu_2$$

$$Q = \begin{pmatrix} r \\ u \end{pmatrix}^T \begin{pmatrix} S^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} r \\ u \end{pmatrix}$$

$$= \underbrace{r^T S^{-1} r}_{\text{residual}} + \underbrace{u^T \Sigma_{22}^{-1} u}_{\text{observed deviation}}$$

For fixed x_2, the observed term is constant in x_1; the residual term varies through $r = x_1 - m(x_2)$.

The same elimination also splits the determinant

Covariance identity Elimination

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

$$\det T = \det I_{d_1} \det I_{d_2} = 1$$

T is triangular, so its determinant is 1; determinants multiply

$$\det\left(T\Sigma T^T\right) = \det T \cdot \det \Sigma \cdot \det T^T = \det \Sigma$$

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$$\det\left(T\Sigma T^T\right) = \det T \cdot \det \Sigma \cdot \det T^T = \det \Sigma$$

$$\det \Sigma = \det S \cdot \det \Sigma_{22}$$

> The elimination preserves the determinant and exposes exactly the two covariance determinants needed for Gaussian normalization.

Dimensions and determinants split the normalizing factor

Coefficient Full density and block dimensions

$$C = (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}, \quad d = d_1 + d_2$$

$$C = (2\pi)^{-\frac{d_1+d_2}{2}} (\det S \cdot \det \Sigma_{22})^{-\frac{1}{2}}$$

Dimensions and determinants split the normalizing factor

Coefficient Full density and block dimensions

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$$C = (2\pi)^{-\frac{d_1+d_2}{2}} (\det S \cdot \det \Sigma_{22})^{-\frac{1}{2}}$$

Split the powers and positive square roots

$$C = \underbrace{\frac{1}{(2\pi)^{\frac{d_1}{2}} \sqrt{\det S}}}_{\text{residual factor}} \cdot \underbrace{\frac{1}{(2\pi)^{\frac{d_2}{2}} \sqrt{\det \Sigma_{22}}}}_{\text{observed factor}}$$

Each coefficient has the dimension and covariance determinant of its Gaussian density; positivity licenses the positive square roots.

Both parts combine into normalized Gaussian factors

Quadratic Quadratic split

$$Q = r^T S^{-1} r + u^T \Sigma_{22}^{-1} u, \quad u = x_2 - \mu_2, \quad r = x_1 - m(x_2)$$

Substitute the separated quadratic and coefficient in $C \exp\left(-\frac{Q}{2}\right)$

$$p(x_1, x_2) = \frac{\exp\left[-\frac{1}{2} r^T S^{-1} r\right]}{(2\pi)^{\frac{d_1}{2}} \sqrt{\det S}} \cdot \frac{\exp\left[-\frac{1}{2} u^T \Sigma_{22}^{-1} u\right]}{(2\pi)^{\frac{d_2}{2}} \sqrt{\det \Sigma_{22}}}$$

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$$p(x_1, x_2) = \mathcal{N}(x_1 | m(x_2), S) \mathcal{N}(x_2 | \mu_2, \Sigma_{22})$$

For each fixed x_2, the first factor is a normalized density in x_1. Its mean still depends on the observation.

Integrating out the residual identifies the marginal

Joint density Previous slide

$$p(x_1, x_2) = \mathcal{N}(x_1 | m(x_2), S)\mathcal{N}(x_2 | \mu_2, \Sigma_{22})$$

Hold x_2 fixed and integrate with respect to x_1

$$p(x_2) = \int_{\mathbb{R}^{d_1}} p(x_1, x_2)\, \mathrm{d}x_1$$

$$= \mathcal{N}(x_2 | \mu_2, \Sigma_{22}) \underbrace{\int_{\mathbb{R}^{d_1}} \mathcal{N}(x_1 | m(x_2), S)\, \mathrm{d}x_1}_{=1}$$

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$$p(x_2) = \mathcal{N}(x_2 | \mu_2, \Sigma_{22}) > 0$$

The factor that depends only on the observation is its marginal density. Its strict positivity makes the density ratio available at every x_2.

Dividing by the marginal gives the conditional Gaussian

Cancel the positive observed-block density

$$p(x_1|x_2) = \frac{p(x_1,x_2)}{p(x_2)} = \frac{\mathcal{N}(x_1|m(x_2),S)\mathcal{N}(x_2|\mu_2,\Sigma_{22})}{\mathcal{N}(x_2|\mu_2,\Sigma_{22})} = \mathcal{N}(x_1|m(x_2),S)$$

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$$m(x_2) = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

> The observed deviation shifts the conditional mean. The conditional covariance is the residual covariance and does not depend on the observed value.

Zero cross-covariance removes the observation

$$\Sigma_{12} = 0 \quad \Rightarrow \quad B = 0, \quad m(x_2) = \mu_1, \quad S = \Sigma_{11}$$

$$p(x_1|x_2) = \mathcal{N}(x_1|\mu_1, \Sigma_{11}) = p(x_1)$$

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$$p(x_1|x_2) = \mathcal{N}(x_1|\mu_1, \Sigma_{11}) = p(x_1)$$

$$p(x_1, x_2) = p(x_1)p(x_2) \quad \Rightarrow \quad X_1 \perp X_2$$

For a joint Gaussian, zero cross-covariance makes observations of one block uninformative about the other.